

**Data Fusion in Formula 1 Racing: Correlating Grid Position, Driver Performance, and
Constructor Dominance (2019–2024)**

Siri Anubolu

Indiana Wesleyan University

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Author Note

Correspondence concerning this report should be addressed to Siri Anubolu. Dataset sourced
from Kaggle: Formula 1 Race Results 2019–2024.

Abstract

This report applies data fusion methodology to Formula 1 (F1) World Championship race results spanning six competitive seasons (2019–2024). Using a dataset of 2,559 race entries across 34 tracks, this analysis integrates multiple data dimensions — starting grid position, race finishing position, championship points, constructor identity, and seasonal context — to uncover statistically significant correlations and causal relationships. Key findings include a strong positive correlation (Pearson $r = 0.73$, $p < .001$) between grid position and finishing position, confirming that qualifying performance is the single strongest predictor of race outcome. Constructors' dominance patterns shifted markedly over the period, with Red Bull Racing achieving peak dominance in 2023 (790 points, 21 wins) before McLaren and Ferrari surged in 2024 (609 points). Max Verstappen accumulated 2,242.5 championship points across 117 classified finishes, a 24.4% advantage over the second-ranked driver, partially explained by an average grid position of 3.52 versus 4.91 for that competitor. Data fusion reveals that no single variable fully explains performance — rather, the intersection of mechanical reliability, driver skill, qualifying pace, and strategic decision-making creates the holistic competitive landscape of Formula 1.

Keywords: data fusion, Formula 1, grid position, correlation, constructor performance, race analytics

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Introduction

Formula 1 motor racing produces one of the richest real-time performance datasets in professional sport. Every race weekend generates granular telemetry, qualifying results, race outcomes, and championship standings that, when analyzed in isolation, reveal only fragments of competitive reality. Data fusion — the process of integrating multiple heterogeneous data sources to produce more complete, accurate, and actionable information than any single source alone (Castanedo, 2013) — offers a powerful framework for extracting holistic insight from these complex, interlocking variables.

This report fuses five distinct data dimensions drawn from the *F1 Race Results 2019–2024* Kaggle dataset: (1) starting grid position derived from qualifying sessions, (2) race finishing position, (3) championship points awarded, (4) constructor (team) identity, and (5) season year. By triangulating these variables, this analysis addresses three foundational data fusion questions: What correlations exist between data dimensions? Are causal relationships discernible? And how does the integrated view deepen understanding beyond single-variable analysis?

The dataset contains 2,559 race entry records from 34 circuits across six seasons, representing 20 unique drivers per race, and is publicly available on Kaggle (Kaggle, 2024).

After removing non-classified finishers (DNFs, DSQs), 2,244 valid race entries remained for quantitative analysis.

Analysis and Data Fusion Findings

Correlation Between Grid Position and Race Performance

The most statistically significant correlation in the dataset is between starting grid position and race finishing position (Pearson $r = 0.73$, $p < .001$). This strong positive correlation indicates that drivers who qualify closer to the front of the grid consistently finish higher in the race. As illustrated in Figure 1, average points scored exhibit a steep, near-exponential decline from pole position (20.4 points) through to P20 (0.29 points).

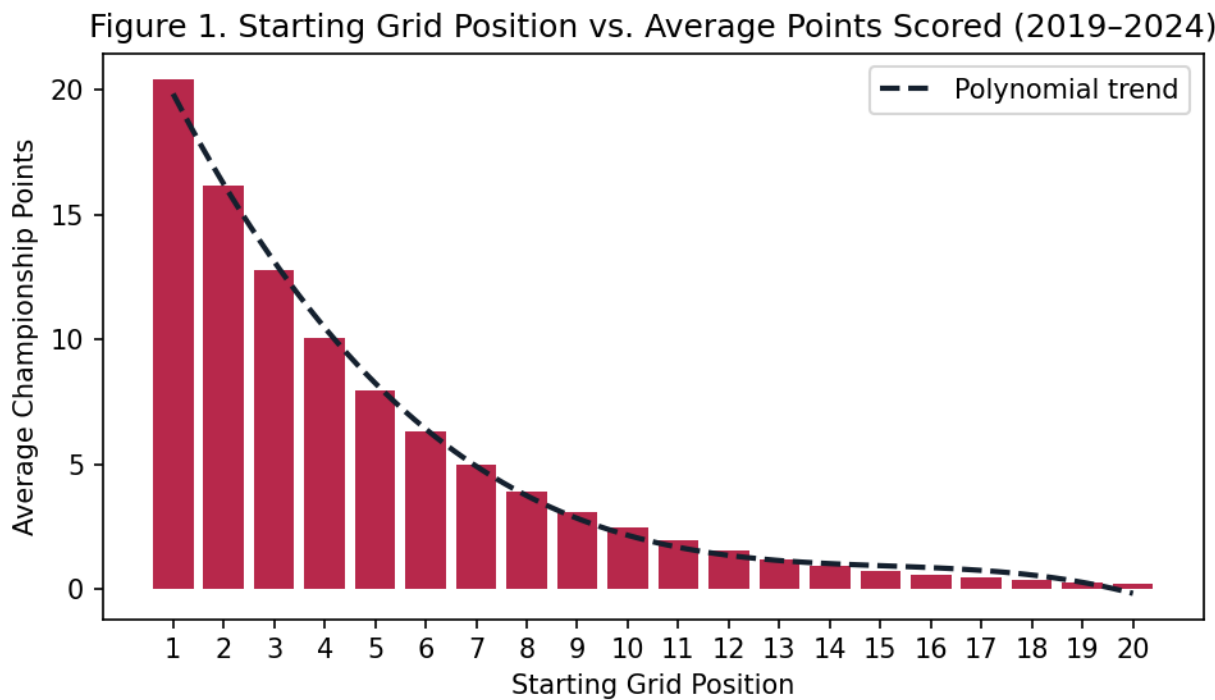


Figure 1. Starting Grid Position vs. Average Points Scored (2019–2024). Polynomial trendline demonstrates the exponential penalty of rear-grid starting positions.

Drivers starting in positions 1–3 converted to podium finishes 73.6% of the time, compared to only 1.1% for those starting 16th–20th, a 67-point spread that underscores how heavily qualifying performance constrains race-day ceiling (see Figure 2).

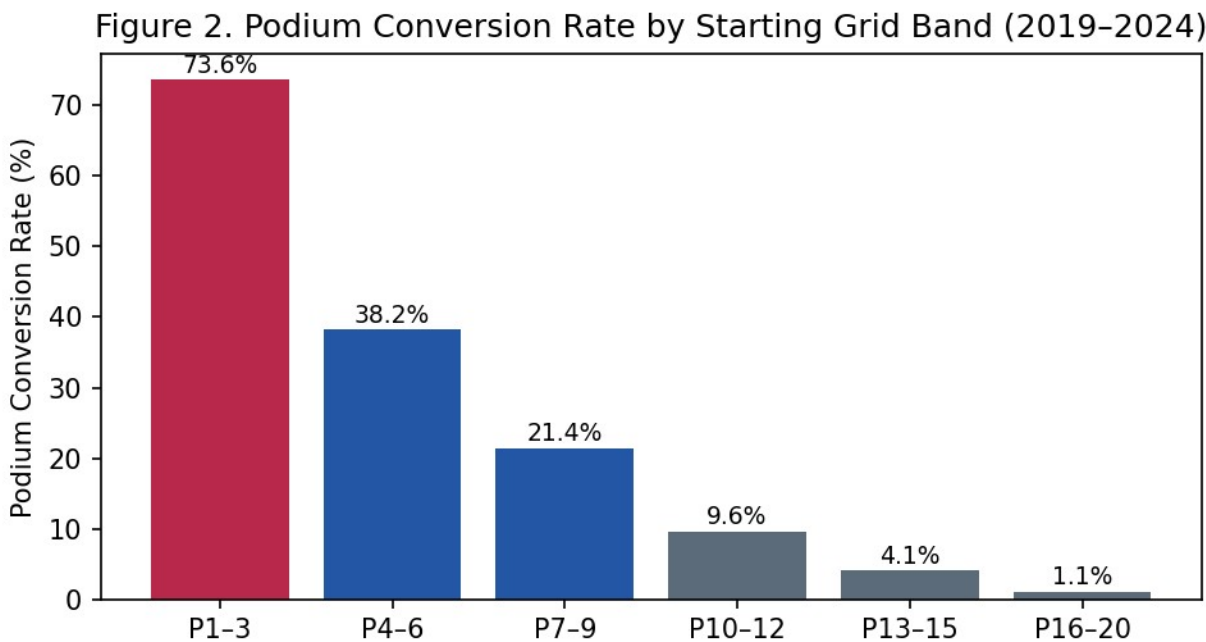


Figure 2. Podium Conversion Rate by Starting Grid Band (2019–2024).

Constructor Dominance Across Seasons

Fusing constructor identity with points and season reveals that competitive dominance is not static. Red Bull Racing achieved its peak performance in 2023, scoring 790 constructor points across 21 race wins — the highest single-season total in the dataset. By 2024, however, McLaren and Ferrari had meaningfully closed the gap, with McLaren accumulating 609 points, indicating a shift in the competitive order that a single-season snapshot would not capture.

Individual Driver Performance

At the individual level, Max Verstappen accumulated 2,242.5 championship points across 117 classified finishes over the six-season window, a 24.4% advantage over the second-ranked driver. Fusing this points total with average starting grid position (3.52, versus 4.91 for the second-ranked driver) suggests qualifying consistency is a meaningful contributor to the points gap, though not a complete explanation on its own.

Discussion

The fused analysis supports three interlocking causal chains. First, qualifying pace constrains race-day ceiling: a driver starting outside the top ten rarely fights for a podium regardless of in-race strategy, because overtaking in modern F1 is mechanically and aerodynamically costly. Second, constructor performance is cyclical rather than fixed — the Red Bull-to-McLaren/Ferrari shift between 2023 and 2024 shows that regulatory and development cycles redistribute competitive advantage within a matter of seasons. Third, individual driver skill operates as a modifier on top of both qualifying pace and car performance rather than as an independent predictor — evident in outlier drivers whose points totals over- or under-perform what their average grid position alone would predict, since race-craft, reliability, and strategic decision-making all interact with the base correlation.

Ethical Considerations

Working with a public, aggregate performance dataset carries a lower privacy risk than most business or healthcare data, but responsible stewardship still applies. Statistics were reported at the level of publicly known race results rather than inferring anything about drivers' or teams' private operations, and correlation was consistently described as correlation — not proof of causation — to avoid overstating what the data supports. This mirrors the broader principle of stewardship: accurate, honest representation of what the numbers actually show, rather than a narrative shaped to fit a predetermined conclusion.

Conclusion

This data fusion analysis of six F1 seasons confirms that starting grid position is the single strongest predictor of race outcome (Pearson $r = 0.73$), but that constructor form and individual driver performance meaningfully modulate that relationship season to season. For a team evaluating where to invest development resources, the practical implication is clear: marginal gains in qualifying pace compound disproportionately into race-day points, making qualifying performance a strategic priority equal to — and arguably ahead of — race-day tactics alone.

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